Infosys System Engineer: Mock Test — Paper 4

Total questions: 60 | **Duration:** 120 minutes | **Sectional cut-offs apply**

Section	Questions	Time
1. Reasoning Ability	Q1–15	25 min
2. Technical Ability (Quant)	Q16–25	35 min
3. Verbal Ability	Q26–45	20 min
4. Pseudo Code	Q46–50	10 min
5. Numeric Puzzle Solving	Q51–54	10 min
6. English Grammar	Q55–59	10 min
7. English Writing	Q60	10 min

SECTION 1: Reasoning Ability (15 Q, 25 min)

Q1. Statements: Some pens are pencils. All pencils are erasers. No eraser is a sharpener.

Conclusions: I. No pencil is a sharpener. II. Some pens are erasers. III. Some erasers are not pens. IV. Either some pens are sharpeners or no pens are sharpeners.

(a) Only I and II follow (b) I, II and IV follow (c) All follow (d) Only IV follows

Q2. Statements: Only members are allowed entry. Some members are officers. All officers are trained.

Conclusions: I. Some trained persons are members. II. All those allowed entry are trained. III. Some members are trained.

(a) I and III only (b) II and III only (c) All follow (d) None follows

Q3. *Data Sufficiency.* What is the value of x?

I. $x^2 - 5x + 6 = 0$

II. $x > 2$

(a) Statement I alone is sufficient, but II alone is not

(b) Statement II alone is sufficient, but I alone is not

(c) Both statements together are sufficient, but neither alone is

(d) Even both statements together are not sufficient

Q4. *Data Sufficiency.* In triangle ABC, what is the measure of angle B?

I. Angle A = 50°, and the triangle is isosceles with AB = AC.

II. Angle C = 65°.

(a) Statement I alone is sufficient, but II alone is not

(b) Statement II alone is sufficient, but I alone is not

(c) Both statements together are sufficient, but neither alone is

(d) Even both statements together are not sufficient

Q5. In a code, each letter is first replaced by its position in the alphabet, and that number is then doubled and reduced by 1. Using this rule, CAT is coded as 5-1-39. What is the code for DOG?

(a) 7-29-13 (b) 8-29-13 (c) 7-30-13 (d) 7-29-14

Q6. In a code: "alpha beta gamma delta" = "zx qr mn kl", "beta gamma epsilon" = "qr mn vw", "gamma delta zeta" = "mn kl tu", "epsilon zeta eta" = "vw tu yz", "alpha eta theta" = "zx yz op". What is the code for "theta beta zeta"?

(a) op qr tu (b) op qr vw (c) zx qr tu (d) op mn tu

Q7. A cube of side 6 units is painted red on all faces and then cut into 1-unit cubes. How many of the small cubes have at least 2 painted faces?

(a) 56 (b) 48 (c) 64 (d) 60

Q8. How many squares (of all sizes) are there in a 4×4 grid of unit squares?

(a) 30 (b) 25 (c) 36 (d) 20

Q9. P is the father of Q. Q is the brother of R. S is the mother of R. T is the husband of S. How is P related to T?

(a) Father (b) P and T are the same person (c) Brother-in-law (d) Cannot be determined

Q10. If A + B means A is the father of B, A − B means A is the mother of B, and A × B means A is the son of B, which expression shows that M is the maternal grandfather of N?

(a) M + Q − N (b) M − Q + N (c) Q + M − N (d) M × Q − N

Q11. A man walks 14 m East, then 20 m North, then 5 m West, then 8 m South. How far is he from the starting point, and in which direction?

(a) 15 m, North-East (b) 15 m, North-West (c) 13 m, North-East (d) 17 m, North-East

Q12. A ship sails 10 km from port on a bearing of 030° (N30°E), then sails 10 km on a bearing of 150° (S30°E). What is the ship's final position relative to the port?

(a) 10 km due East (bearing 090°) (b) 10 km due North (c) 17.3 km, bearing 060° (d) 10 km, bearing 120°

Q13. Find the next term: 2, 5, 12, 27, 58, ?

(a) 121 (b) 125 (c) 115 (d) 118

Q14. Find the next term: 1, 3, 7, 17, 41, ?

(a) 99 (b) 95 (c) 101 (d) 89

Q15. Find the next term: Y, V, R, M, ?

(a) G (b) H (c) F (d) I

SECTION 2: Technical Ability, Quantitative (10 Q, 35 min)

Q16. In how many ways can 6 men and 4 women be seated at a round table such that no two women sit together?

(a) 43,200 (b) 86,400 (c) 21,600 (d) 14,400

Q17. In how many ways can the letters of the word ORANGE be arranged so that the vowels occupy only the odd positions?

(a) 36 (b) 72 (c) 18 (d) 144

Q18. Three machines A, B and C produce 25%, 35% and 40% of a factory's output respectively, with defect rates of 5%, 4% and 2%. An item is picked at random and found defective. What is the probability it came from machine C?

(a) 16/69 (b) 8/69 (c) 23/69 (d) 25/69

Q19. 5 boys and 3 girls are seated randomly in a row of 8 chairs. What is the probability that all 3 girls sit together?

(a) 3/28 (b) 1/8 (c) 3/56 (d) 5/28

Q20. A trader buys 80 kg of rice at ₹25/kg. He sells 50 kg at a 20% profit and wants his overall profit on all 80 kg to be 15%. At what price per kg must he sell the remaining 30 kg?

(a) ₹26.67 (b) ₹25 (c) ₹28 (d) ₹24

Q21. A, B and C can complete a work alone in 20, 30 and 60 days respectively. All three start together. After 4 days, B leaves; after 3 more days, C also leaves, and A finishes the rest alone. In how many total days is the work completed?

(a) 15 days (b) 14 days (c) 16 days (d) 13 days

Q22. Two trains, 300 km apart, move towards each other on the same track. Train A departs at 8:00 AM at 60 km/h; Train B departs at 8:30 AM at 90 km/h. At what time do they meet?

(a) 10:18 AM (b) 10:12 AM (c) 10:06 AM (d) 10:24 AM

Q23. In a two-candidate election, the winner received 60% of votes cast and won by 4,800 votes. If 20% of registered voters did not vote and there were no invalid votes, how many voters were registered?

(a) 30,000 (b) 28,000 (c) 32,000 (d) 24,000

Q24. Find the compound interest on ₹12,000 for 1.5 years at 10% per annum, compounded annually (interest for the fractional half-year is simple interest on the amount at the end of the first year).

(a) ₹3,860 (b) ₹3,800 (c) ₹3,960 (d) ₹3,720

Q25. The compound interest on a sum for 2 years is ₹1,050, and for 3 years is ₹1,655, at the same annual rate. The simple interest on the same sum for 2 years, at the same rate, is ₹1,000. Find the rate of interest per annum.

(a) 10% (b) 12% (c) 8% (d) 15%

SECTION 3: Verbal Ability (20 Q, 20 min)

Read the passage and answer Q26–30.

> *Distributed systems theory offers a deceptively simple trilemma: a system cannot simultaneously guarantee consistency, availability and partition tolerance, and must therefore sacrifice one when a network partition occurs. In practice, however, the theorem is more often invoked than correctly applied, because partition tolerance is not actually optional for any system whose nodes communicate over an unreliable network, which is to say nearly all distributed systems built today. The real choice, once partition tolerance is accepted as a structural necessity rather than a design preference, is between consistency and availability during the partition itself, and this choice need not be uniform across an entire system. A payment ledger might reasonably refuse writes during a partition to preserve consistency, accepting temporary unavailability as the lesser cost, while a product catalogue serving read traffic might instead continue answering requests with potentially stale data, judging availability the more valuable property for that workload. Treating the CAP theorem as prescribing a single, system-wide*

> *choice, rather than a per-operation or per-service one, is therefore a common and consequential misreading.*

Q26. According to the passage, the CAP theorem is frequently misapplied because:
(a) Partition tolerance is wrongly treated as an optional design choice when it is effectively mandatory
(b) Consistency is always more important than availability
(c) The theorem does not apply to payment systems
(d) Network partitions rarely occur in practice

Q27. The payment-ledger versus product-catalogue example illustrates that:
(a) Payment systems should never use distributed architecture
(b) The consistency-availability trade-off can be made differently for different parts of the same overall system
(c) Product catalogues are inherently less important than payment systems
(d) Availability should always be prioritised over consistency

Q28. The phrase "a common and consequential misreading" refers to:
(a) Confusing consistency with availability
(b) Treating the CAP choice as one uniform, system-wide decision rather than one that can vary per service
(c) Believing that partition tolerance can be sacrificed
(d) Assuming all networks are reliable

Q29. Which of the following would the author most likely agree with?
(a) A single distributed system can legitimately favour consistency for one function and availability for another
(b) The CAP theorem is largely irrelevant to real-world system design
(c) Partition tolerance should be sacrificed whenever possible
(d) All read-heavy services must prioritise consistency

Q30. Based on the passage, why is partition tolerance called a "structural necessity"?
(a) Because regulations require it
(b) Because any system communicating over an unreliable network cannot realistically avoid the possibility of partitions
(c) Because consistency is always more important
(d) Because availability is always more important

Sentence Correction (Q31–35).

Q31. It is imperative that every deployment **is tested** thoroughly before being pushed to production.
(a) No improvement (b) be tested (c) was tested (d) will be tested

Q32. Rarely **a system architect gets** the chance to redesign a platform from scratch.
(a) No improvement (b) does a system architect get (c) a system architect does get (d) gets a system architect

Q33. The report's conclusion, **that many have questioned it**, was based on incomplete data.
(a) No improvement (b) which many have questioned (c) that many have questioned (d) many have questioned which

Q34. The budget overrun was attributed **to the fact of poor estimation practices** rather than external factors.

(a) No improvement (b) to poor estimation practices (c) to the fact that poor estimation practices (d) because of poor estimation practices

Q35. Being that the deadline was moved up, the team had to prioritise ruthlessly.

(a) No improvement (b) Given that the deadline was moved up (c) Being the deadline was moved up (d) Because of the deadline was moved up

Para-jumbles (Q36–38).

Q36.

P: Yet this very redundancy, when poorly managed, can itself become a source of inconsistency.

Q: Distributed databases often replicate data across multiple nodes to improve fault tolerance.

R: Conflict-resolution strategies, such as vector clocks, were developed to reconcile such divergent copies.

S: If two replicas are updated independently before synchronising, they may end up holding conflicting values.

(a) QPSR (b) QSPR (c) PQSR (d) QRPS

Q37.

P: It instead reflects a deliberate trade-off between latency and durability that every system designer must make explicitly.

Q: Writing data to disk before acknowledging a request guarantees durability but adds latency.

R: Skipping this step and acknowledging immediately improves responsiveness at the risk of data loss during a crash.

S: Neither approach is objectively superior, and the choice is not a matter of engineering carelessness.

(a) QRSP (b) QSRP (c) RQSP (d) QRPS

Q38.

P: This made debugging nearly impossible, since no single node had a complete picture of the request's path.

Q: Early microservice deployments often lacked any form of distributed tracing.

R: Modern observability platforms now stitch together logs from every service a request touches.

S: A single user request might traverse a dozen services before returning a response.

(a) QSPR (b) QPSR (c) SQPR (d) QSRP

Vocabulary, Synonyms and Antonyms (Q39–45).

Q39. The auditor's findings were **unequivocal**, leaving no room for alternative interpretation. "Unequivocal" means:

(a) ambiguous (b) leaving no doubt; completely clear (c) tentative (d) biased

Q40. Critics dismissed the proposal as **specious** — appealing on the surface but fundamentally flawed. "Specious" means:

(a) genuinely sound (b) superficially plausible but actually wrong (c) deeply researched (d) widely accepted

Q41. Fill in the blank: Despite years of effort, the two departments remained _____ in their approaches, unable to find common ground.

(a) aligned (b) intransigent (c) harmonious (d) cooperative

Q42. Synonym of **CIRCUMSPECT**:

(a) reckless (b) cautious and careful (c) indifferent (d) impulsive

Q43. Synonym of **OBFUSCATORY**:

(a) clarifying (b) deliberately confusing (c) simplifying (d) transparent

Q44. Antonym of **PARSIMONIOUS**:

(a) frugal (b) stingy (c) lavish (d) economical

Q45. Antonym of **GARRULOUS**:

(a) talkative (b) chatty (c) taciturn (d) verbose

SECTION 4: Pseudo Code (5 Q, 10 min)

Q46. What is printed? (`%` is the modulo operator)

```
int gcd(int a, int b) {
    if (b == 0) return a;
    return gcd(b, a % b);
}
print(gcd(252, 105))
```

(a) 21 (b) 42 (c) 7 (d) 14

Q47. What is printed? (^=XOR, |=OR, &=AND, >>=right shift)

```
a = 13; b = 25;
c = (a ^ b) & (a | b);
d = c >> 2;
print(c + d)
```

(a) 25 (b) 29 (c) 20 (d) 33

Q48. How many swaps are performed in total?

```
arr = [5, 2, 8, 1, 9, 3]
swaps = 0
for i = 0 to 4
    for j = 0 to 4 - i
        if (arr[j] > arr[j+1])
            temp = arr[j]
            arr[j] = arr[j+1]
            arr[j+1] = temp
            swaps = swaps + 1
        end if
    end for
end for
print(swaps)
```

(a) 7 (b) 8 (c) 6 (d) 9

Q49. This algorithm reverses the first `k` elements of a queue using a stack. What is printed?

```
queue = [1, 2, 3, 4, 5]
stack = []
k = 3
for i = 1 to k
    push(stack, dequeue(queue))
end for
while (stack is not empty)
    enqueue(queue, pop(stack))
end while
for i = 1 to (5 - k)
    enqueue(queue, dequeue(queue))
end for
print(queue)
```

(a) [3, 2, 1, 4, 5] (b) [1, 2, 3, 4, 5] (c) [5, 4, 3, 2, 1] (d) [3, 2, 1, 5, 4]

Q50. This algorithm moves all non-zero elements to the front, preserving order, and fills the rest with zeroes. What is the final `arr`?

```
arr = [3, 0, 4, 0, 5, 0, 2]
pos = 0
for i = 0 to 6
    if (arr[i] != 0)
        arr[pos] = arr[i]
        pos = pos + 1
    end if
end for
while (pos <= 6)
    arr[pos] = 0
    pos = pos + 1
end while
print(arr)
```

(a) [3, 4, 5, 2, 0, 0, 0] (b) [3, 0, 4, 0, 5, 0, 2] (c) [3, 4, 5, 2, 0, 0, 2] (d) [0, 0, 0, 3, 4, 5, 2]

SECTION 5: Numeric Puzzle Solving (4 Q, 10 min)

Q51. *Matchstick.* A 3×3 grid of unit squares (9 small squares) is built using 24 matchsticks. Counting squares of every size (1×1, 2×2 and 3×3 — 14 squares in total), what is the minimum number of matchsticks that must be removed so that no complete square of any size remains?

(a) 4 (b) 5 (c) 6 (d) 7

Q52. *Circular wheel.* Each wheel has numbers at Top, Right, Bottom, Left and a Centre number formed by the same rule.

Wheel 1: T = 4, R = 5, B = 6, L = 3, Centre = 39

Wheel 2: T = 7, R = 2, B = 8, L = 5, Centre = 66

Wheel 3: T = 6, R = 4, B = 9, L = 2, Centre = ?

(a) 58 (b) 62 (c) 66 (d) 54

Q53. *Matrix logic.* Find the missing number.

2	3	4
6	8	10
12	15	?

(a) 16 (b) 18 (c) 20 (d) 21

Q54. *Pattern.* Find the next term: 1, 5, 14, 30, 55, ?

(a) 91 (b) 85 (c) 96 (d) 90

SECTION 6: English Grammar (5 Q, 10 min)

Q55. _____ the system crash occurred during peak hours, the financial losses would have been catastrophic.

(a) If (b) Had (c) Should (d) Unless

Q56. The engineer explained that the server _____ down due to a memory leak that _____ undetected for weeks.

(a) went, had gone (b) had gone, went (c) goes, had gone (d) went, has gone

Q57. The team is responsible _____ delivering the project _____ budget and _____ the agreed deadline.
(a) for, within, by (b) of, under, before (c) for, under, within (d) to, in, at

Q58. *Error spotting.* "The new system is not only faster than the old one (A) / but also requires (B) / less maintenance, making it (C) / a more cost-effective to operate. (D)"
(a) A (b) B (c) C (d) D

Q59. _____ information technology has transformed _____ way we work, though _____ access to _____ internet remains unequal across regions.
(a) -, the, -, the (b) The, the, an, the (c) -, the, the, the (d) An, the, -, an

SECTION 7: English Writing (1 Q, 10 min)

Q60. *Write a short technical essay/paragraph of 150–180 words.*
A legacy monolithic application your team supports is becoming difficult to maintain and scale. Explain the key risks of migrating it to a microservices architecture versus keeping it monolithic but modularising the codebase internally, and recommend one approach for a team of 8 engineers with limited DevOps experience, justifying your choice.

Evaluation rubric (10 marks): Technical accuracy of the comparison (3) | Clarity and soundness of the recommendation given the team's constraints (3) | Structure and coherence (2) | Grammar and vocabulary (1) | Word limit adherence (1)

ANSWER KEY WITH EXPLANATIONS

Section 1: Reasoning

1. **(b)** Pencils⊆erasers and erasers∩sharpeners=∅, so pencils∩sharpeners=∅ (I follows). Some pens are pencils, and pencils⊆erasers, so some pens are erasers (II follows). III isn't guaranteed (no information ensures an eraser outside the pen-pencil overlap exists). For IV: we know the pen-pencils aren't sharpeners, but nothing is said about pens that aren't pencils, so neither "some pens are sharpeners" nor "no pens are sharpeners" can be individually proven — exactly the condition for the standard either-or rule, so IV follows.

2. **(a)** "Only members are allowed entry" means allowed-entry⊆members. Some members are officers, and officers⊆trained, so those members are trained — this directly gives "some members are trained" (III) and, by simple conversion, "some trained are members" (I). But since only *some* members are officers/trained, we cannot say *all* those allowed entry are trained (II fails, as non-officer members may not be trained).

3. **(c)** I alone: $x^2-5x+6=0$ factors to $(x-2)(x-3)=0$, giving $x=2$ or $x=3$ — two solutions, insufficient. II alone: $x>2$ alone is insufficient. Together: only $x=3$ satisfies both $x>2$ and the equation — sufficient together.

4. **(a)** I alone: AB=AC means the angles opposite them are equal, so angle B = angle C; with angle A=50°, B+C=130°, and B=C gives B=65° — sufficient. II alone gives only one angle with no other relation — insufficient.

5. **(a)** D→4→2(4)−1=7. O→15→2(15)−1=29. G→7→2(7)−1=13. Code: **7-29-13**.

6. **(a)** Comparing statements pairwise for common words and codes: shines=pu, gamma=mn, bright=mo (from statements 1&3), giving delta=kl (from statements 1&3's remaining elements), light=ta, zeta=tu, epsilon=vw, eta=yz, alpha=zx, theta=op. So "theta beta zeta" = **op qr tu**.

7. **(a)** For an n×n×n cube, cubes with exactly 2 painted faces (edges) = $12(n-2)$; with exactly 3 (corners) = 8. For n=6: edges=12×4=48, corners=8, so at least 2 = 48+8 = **56**.

8. **(a)** Squares of size k in an n×n grid = $(n-k+1)^2$. For n=4: k=1→16, k=2→9, k=3→4, k=4→1. Total = 16+9+4+1 = **30**.

9. **(b)** Q and R are siblings (Q is R's brother), so S (R's mother) is also Q's mother. P is Q's father, so P and S are Q's parents — meaning P is S's husband. Since T is defined as S's husband, **P and T are the same person**.

10. **(a)** M+Q: M is father of Q. Q−N: Q is mother of N. So M is father of Q, who is N's mother — M is N's **maternal grandfather**.

11. **(a)** Net East-West = 14−5 = 9 m East; net North-South = 20−8 = 12 m North. Distance = $\sqrt{(9^2+12^2)} = \sqrt{225}$ = **15 m**, direction **North-East**.

12. **(a)** Leg 1 (bearing 030°, 10 km): East = 10 sin30° = 5, North = 10 cos30° = $5\sqrt{3}$. Leg 2 (bearing 150°, 10 km): East = 10 sin150° = 5, North = 10 cos150° = $-5\sqrt{3}$. Totals: East = 5+5 = 10, North = $5\sqrt{3}-5\sqrt{3}=0$. Final position: **10 km due East (bearing 090°)**.

13. **(a)** Differences are 3, 7, 15, 31 — each is 2^n-1 (n=2,3,4,5); next difference is $2^6-1=63$. 58+63 = **121**.

14. **(a)** Each term follows a(n) = 2×a(n−1) + a(n−2): 2×17+7=41 ✓; next = 2×41+17 = **99**.

15. **(a)** Gaps between letters increase by 1 each time: −3, −4, −5, next −6. M(13)−6 = 7 = **G**.

Section 2: Quant

16. **(a)** Seat 6 men in a circle: (6−1)! = 120 ways. This creates 6 gaps; choose and arrange 4 women in 4 of these 6 distinct gaps: P(6,4) = 6×5×4×3 = 360. Total = 120×360 = **43,200**.

17. **(a)** ORANGE has 3 vowels (O,A,E) and 3 consonants (R,N,G), matching the 3 odd and 3 even positions exactly. Vowels in odd positions: 3! = 6. Consonants in even positions: 3! = 6. Total = 6×6 = **36**.

18. **(a)** P(D) = 0.25×0.05 + 0.35×0.04 + 0.40×0.02 = 0.0125+0.014+0.008 = 0.0345. P(C|D) = (0.40×0.02)/0.0345 = 0.008/0.0345 = **16/69**.

19. **(a)** Treating the 3 girls as one block: favourable arrangements = 6!×3! = 720×6 = 4,320. Total arrangements = 8! = 40,320. Probability = 4320/40320 = **3/28**.

20. **(a)** Total CP = 80×25 = ₹2,000; target total SP (15% profit) = ₹2,300. SP from the first 50 kg (cost ₹1,250, at 20% profit) = ₹1,500. Remaining SP needed = 2300−1500 = ₹800 for 30 kg → ₹800/30 ≈ **₹26.67/kg**.

21. **(a)** Rates (LCM 60): A=3/60, B=2/60, C=1/60 per day. First 4 days (all three): 4×6/60=24/60 done. Next 3 days (A+C only): 3×4/60=12/60 done — total 36/60=3/5 done, 2/5=24/60 remains. A alone at 3/60/day takes 24/60÷3/60 = 8 more days. Total = 4+3+8 = **15 days**.

22. **(a)** From 8:00 to 8:30, only A moves: 60×0.5=30 km, leaving 270 km. From 8:30, combined speed = 150 km/h; time = 270/150 = 1.8 hr = 1 hr 48 min. Meeting time = 8:30 AM + 1:48 = **10:18 AM**.

23. **(a)** Winner's margin (60%−40%=20% of votes cast) = 4,800 → votes cast = 4800/0.20 = 24,000. Since this is 80% of registered voters: registered = 24000/0.80 = **30,000**.

24. **(a)** Year 1: 12000×1.10 = 13,200 (CI so far = 1,200). For the next half-year, SI on 13,200 at 10% for 0.5 yr = 660. Total amount = 13,860; total CI = 13860−12000 = **₹3,860**.

25. **(a)** 3rd-year interest = 1655−1050 = ₹605 — but more directly: $CI_2 - SI_2 = Pr^2$ = 1050−1000 = 50, and SI_2 = 2Pr = 1000 so Pr=500. Dividing, r = 50/500 = 0.10 = **10%** (and P=₹5,000, verified: 5000×1.1^3=6,655, CI_3=1,655 ✓).

Section 3: Verbal

26. **(a)** 27. **(b)** 28. **(b)** 29. **(a)** 30. **(b)**

31. **(b)** "Imperative that" triggers the subjunctive mood, which uses the base form: "be tested".

32. **(b)** "Rarely" at the start of a sentence triggers subject-auxiliary inversion: "does a system architect get".

33. **(b)** The redundant "it" must be dropped, and "which" (not "that") is used after a comma for a non-restrictive clause.

34. **(b)** "The fact of" is redundant and unidiomatic; "attributed to poor estimation practices" is correct and concise.

35. **(b)** "Being that" is non-standard; the correct formal causal connector is "Given that".

36. **(a)** Q→P→S→R. 37. **(a)** Q→R→S→P. 38. **(a)** Q→S→P→R.

39. **(b)** 40. **(b)** 41. **(b)** 42. **(b)** 43. **(b)** 44. **(c)** 45. **(c)**

Section 4: Pseudo Code

46. **(a) 21.** gcd(252,105): 252%105=42 → gcd(105,42): 105%42=21 → gcd(42,21): 42%21=0 → gcd(21,0)=**21**.

47. **(a) 25.** 13=01101, 25=11001. a^b = 10100 = 20. a|b = 11101 = 29. c = 20 & 29 = 10100 = 20. d = 20>>2 = 5. c+d = 20+5 = **25**.

48. **(a) 7.** Full bubble-sort trace: pass i=0 gives 3 swaps (array becomes [2,5,1,8,3,9]); pass i=1 gives 2 more swaps ([2,1,5,3,8,9]); pass i=2 gives 2 more swaps ([1,2,3,5,8,9], now sorted); passes i=3 and i=4 give 0 swaps. Total = 3+2+2 = **7**.

49. **(a)** This is the standard "reverse first k of a queue" algorithm. After moving the first 3 elements through the stack, the queue becomes [4,5,3,2,1]; the final rotation step (dequeue+enqueue twice) moves 4 and 5 to the back, giving **[3,2,1,4,5]** — the first 3 elements correctly reversed.

50. **(a)** Non-zero elements (3,4,5,2) are copied to the front in order, giving positions 0–3; the while loop then zeroes out positions 4,5,6. Final: **[3,4,5,2,0,0,0]**.

Section 5: Numeric Puzzles

51. **(c) 6.** This was verified by exhaustive computer search over all 24 sticks and all 14 squares (nine 1×1, four 2×2, one 3×3): no set of 5 or fewer sticks intersects every square's stick-set, but a 6-stick set exists that does (for example, removing the sticks at positions H(0,0), H(1,0), H(1,1), H(2,1), H(0,2) and V(2,2) breaks all 14 squares). Minimum = **6**.

52. **(b) 62.** Rule: Centre = (T+R)×(B+L) − (T×L + R×B). Wheel 1: (4+5)(6+3)−(4×3+5×6) = 81−42 = 39 ✓. Wheel 2: (7+2)(8+5)−(7×5+2×8) = 117−51 = 66 ✓. Wheel 3: (6+4)(9+2)−(6×2+4×9) = 110−48 = **62**.

53. **(b) 18.** Rule: cell(row r, col c) = r×c + r² (rows and columns numbered from 1). Row 3: (3×1+9, 3×2+9, 3×3+9) = (12, 15, **18**).

54. **(a) 91.** Each term is the sum of squares $1^2+2^2+...+n^2$: 1, 1+4=5, +9=14, +16=30, +25=55, next +36 = **91**.

Section 6: Grammar

55. **(b)** Inverted third conditional (omitting "if"): "Had the crash occurred..." = "If the crash had occurred...".

56. **(a)** The main reported event ("went down") takes simple past; the leak's undetected state, which existed before and up to that event, takes past perfect ("had gone undetected").

57. **(a)** "responsible for" ... "within budget" ... "by the deadline".

58. **(d)** "cost-effective" is an adjective, not a noun, so "a more cost-effective to operate" is ungrammatical — the error is in part (D).

59. **(a)** "Information technology" (general/abstract) takes no article; "the way we work" takes "the" (specific, fixed phrase); "access" (general) takes no article; "the internet" conventionally takes "the".

Section 7: Writing

Use the rubric above. A strong answer explains that microservices offer better independent scalability but introduce operational complexity (service discovery, distributed tracing, orchestration) that a small team with limited DevOps experience may struggle to run reliably, while internal modularisation keeps deployment simple but still improves maintainability. Given the stated constraints (8 engineers, limited DevOps experience), the well-justified recommendation is usually to modularise internally first and revisit microservices once the team's operational maturity grows.

Scoring guide

Award +1 per MCQ (Q1–59) and up to 10 for Q60. This paper is intentionally harder than Papers 1–3; a lower raw score is expected — focus on the reasoning in the explanations, especially the computationally-verified matchstick answer (Q51) and the multi-pass algorithm traces (Q48, Q49).